Python Programmation

CM 6 : Python Libraries

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COMPOSANTE DE L'UNIVERSITÉ DE LORRAINE

Loria

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2 Libraries

3 Environments

4 Modules

What did you learn during last lecture ?
Any questions ?

Last Lecture Doggy bag

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What is a Library

```
import os
import json
import glob

import pandas as pd
from tqdm import tqdm
```

Libraries are the toolboxes of Python

- Contains pre-written code and functions
- Can be installed and added freely
- Designed for specific usecases

An Example: the "os" Library

```
import os

# some calls from the os Library
os.walk()
os.rename()
os.mkdir()
os.exit()
```

"os" is a standard library

- Installed by default with Python
- Other libraries need to be manually added

Most libraries are not Installed by Default

```
import pandas as pd
df = pd.read_csv('data.csv')
```

```
> python my_script.py
Traceback (most recent call last):
  File "my_script.py", line 1, in <module>
    import pandas
ImportError:  No module named pandas
```

How to Install a Library

With pip

```
pip install pandas
```

With uv

```
uv add pandas
```

With conda

```
conda install pandas
```

Installing a List of Libraries

With pip

```
pip install -r requirements.txt
```

With uv

```
uv add -r requirements.txt
```

With conda

```
conda create -file environment.yml
```

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Common Issues in Python

Library versions issues

- One script needs pandas v2.2.3 and the other needs pandas v3.0.6
- You can only have one version of a library

Python versions issues

- You want to use an old script, but it uses Python 2.7
- You can have only one version of Python installed

Solution: create Python virtual environments !

What is a Python Virtual Environment ?

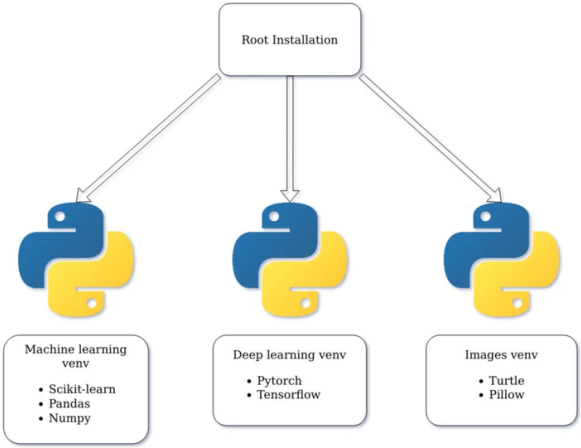
From W3School[1]

> A virtual environment in Python is an isolated environment on your computer, where you can run and test your Python projects.

Python virtual environments (venv) allow you to manage projects which have different Python configurations.

[1] `https://www.w3schools.com/python/python_virtualenv.asp`

How does a venv Works ?



Source: `https://www.python-engineer.com/`

Venv Cheatsheet (python + pip)

How to create a venv

```
> python -m venv venv_name
> source venv_name/bin/activate
```

How to use jupyter-notebook with this venv

```
> pip install -user ipykernel
> python -m ipykernel install -user
```

Venv Cheatsheet (uv)

How to create a venv

```
> cd my_project
> uv init
```

How to use jupyter-notebook with this venv

```
> uv add jupyter
> uv run jupyter notebook
```

Venv Cheatsheet (conda)

How to create a venv

```
> conda create -name venv_name
> conda activate venv_name
```

How to use jupyter-notebook with this venv

```
> conda install jupyter
> jupyter notebook
```

Using the Default environment is BAD

Library compatibilities issues

- Library A and B have a compatibility issue
- You WILL end up re-installing everything

Reproductibility issues

- You want to upload your code with mandatories libraries
- You want to list each library and the version used - as well as their dependencies

Most of the time, you won't even realize there is an issue !

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